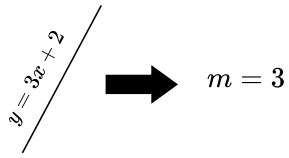
Differentiation

Chapter 12 · Original slide 3

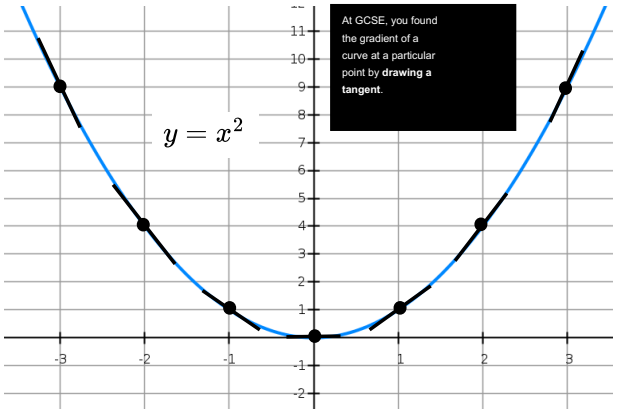
Gradient Function

3

For a straight line, the gradient is **constant**:



However, for a curve **the gradient varies**. We can no longer have a single value for the gradient; **we ideally want an expression in terms of $\mathbf{x}$** that gives us the gradient for any value of x (unsurprisingly known as the **gradient function**).



By looking at the relationship between x and the gradient at that point, can you come up with an expression, in terms of _____ for the gradient?

_____

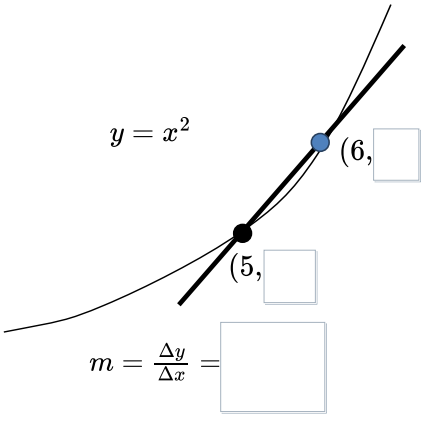
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Finding the Gradient Function

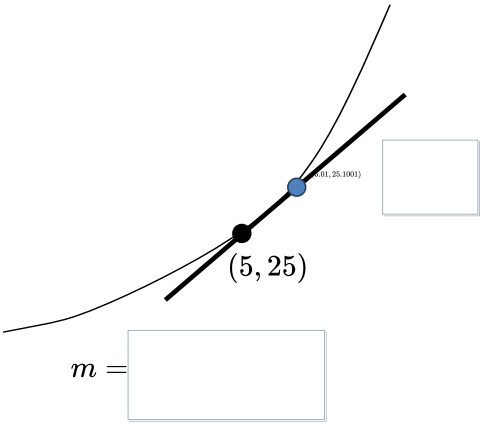
4

The question is then: Is there a method to work out the gradient function without having to draw lots of tangents and hoping that we can spot the rule?

To approximate the gradient on the curve $y = x^2$ when $x = 5$, we could pick a point on the curve just slightly to the right, then find the gradient between the two points:



As the second point gets closer and closer, the gradient becomes a better approximation of the true gradient:



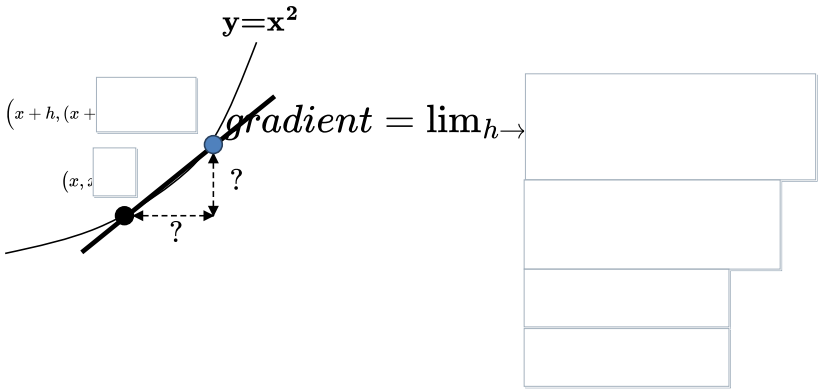
Differentiation

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Finding the Gradient Function

5

This gives us a numerical method to get the gradient **at a particular x**, but doesn't give us the gradient function in general. Let's use exactly the same method, but keep x general, and make the 'small change' (which was previously 0.01) 'h':



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Finding the Gradient Function

6

The gradient function, or derivative, of the curve $y = f(x)$ is written as $f\prime(x)$ or $\frac{dy}{dx}$.

$$f'(x) = \lim_{h\to 0} \frac{f(x+h)-f(x)}{h}$$

The gradient function can be used to find the gradient of the curve for any value of x.

We will soon see an easier/quicker way to differentiate expressions like $y = x^3 - x$ without using 'limits'. But this method, known as **differentiating by first principles**, is now in the A Level syllabus, and you could be tested on it!

Notation Note:

Whether we use $\frac{dy}{dx}$ or $f\prime(x)$ for the gradient function depends on whether we use $y =$ or $f(x) =$ to start with:

"Leibniz's notation"

$$y = x^2 \quad \rightarrow \quad \frac{dy}{dx} = 2x$$
$$f(x) = x^2 \quad \rightarrow \quad f'(x) = 2x$$

"Lagrange's notation"

Advanced Notation Note:

Rather than h for the small change in x, the formal notation is δx. So actually:

$$f'(x) = \lim_{\delta x\to 0} \frac{f(x+\delta x)-f(x)}{\delta x}$$

So we in fact have 3 symbols for "change in"!

- Δx: any change in x (as seen in Chp5: $m = \frac{\Delta y}{\Delta x}$)
- δx: a small change in x
- dx: an infinitesimally small change in x

There's in fact a third way to indicate the gradient function, notation used by Newton: (but not used at A Level)

$$y = x^2 \quad \rightarrow \quad \dot{y} = 2x$$

So the estimated gradient using some point close by was $\frac{\delta y}{\delta x}$, but in the 'limit' as $\delta x \to 0$, $\frac{\delta y}{\delta x} \to \frac{dy}{dx}$

Differentiation

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Example

7

The point A with coordinates $(4, 16)$ lies on the curve with equation $y = x^2$.
At point A the curve has gradient g.
a) Show that $g = \lim_{h \to 0} (8 + h)$
b) Deduce the value of g.

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Test Your Understanding

8

Prove **from first principles** that the derivative of x^4 is $4x^3$.

$f'(x$
$= \lim$
$= \lim$
$= \lim$
$= \lim$
$= 4x$

I's

Differentiation

Chapter 12 · Original slide 9

Exercise 12B – Pearson Pure Y1 page 261-262

9

5 Prove, from first principles, that the derivative of $6x$ is 6.

Reminder:
$f\prime(x)$ and $\frac{dy}{dx}$ both mean the gradient function, also known as the derivative of y.

$$f'(x) = \frac{dy}{dx} = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

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Exercise 12B – Pearson Pure Y1 page 261-262

10

6 Prove, from first principles, that the derivative of $4x^2$ is $8x$.

Reminder:
$f\prime(x)$ and $\frac{dy}{dx}$ both mean the gradient function, also known as the derivative of y.

$$f'(x) = \frac{dy}{dx} = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

Differentiation

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Exercise 12B – Pearson Pure Y1 page 261-262

11

7 $\mathrm{f}(x) = ax^2$, where a is a constant. Prove, from first principles, that $\mathrm{f}'(x) = 2ax$.

Reminder:
$f\prime(x)$ and $\frac{dy}{dx}$ both mean the gradient function, also known as the derivative of y.
$f'(x) = \frac{dy}{dx} = \lim_{h\to 0} \frac{f(x+h)-f(x)}{h}$

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Exercise 12B – Pearson Pure Y1 page 261-262

12

Challenge\quad $\mathrm{f}(x) = \frac{1}{x}$.
a Given that $\mathrm{f}'(x) = \lim_{h\to 0} \frac{\mathrm{f}(x+h)-\mathrm{f}(x)}{h}$, show that $\mathrm{f}'(x) = \lim_{h\to 0} \frac{-1}{x^2+xh}$.
b Deduce that $\mathrm{f}'(x) = -\frac{1}{x^2}$.

Differentiation

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Differentiating x^n

14

Thankfully, there's a quick way to differentiate terms of the form x^n (where n is a constant) with having to use first principles every time:

! If $y = ax^n$ then $\frac{dy}{dx} = nax^{n-1}$ (where a, n are constants)
i.e. multiply by the power and reduce the power by 1

Examples:

$y = x^5 \quad \rightarrow \quad \frac{dy}{dx} =$

$f(x) = x^{\frac{1}{2}} \quad \rightarrow$

$y = 2x^6 \quad \rightarrow$

$f(x) = \frac{x}{x^4} = \quad \rightarrow$

$y = \sqrt{x^6} = \quad \rightarrow$

Why would it be incorrect to say that $y=2^x$ differentiates to $\frac{dy}{dx}=x2^{x-1}$?

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Test Your Understanding

15

1. $y = x^7 \quad \rightarrow \quad \frac{dy}{dx} =$
2. $y = 3x^{10} \quad \rightarrow$
3. $f(x) = \frac{x^{\frac{1}{2}}}{x^2} = \quad \rightarrow$
4. $y = ax^a \quad \rightarrow$
5. $f(x) = \sqrt{49x^7} = \quad \rightarrow \quad f'(x) =$

Differentiation

Chapter 12 · Original slide 16

Differentiating Multiple Terms

16

Differentiate $y = x^2 + 4x + 3$

First thing to note:
If $y = f(x) + g(x)$ then
$\frac{dy}{dx} = f'(x) + g'(x)$
i.e. **differentiate each term individually in a sum/subtraction**.

$\frac{dy}{dx} =$ ☐ ☐ ☐

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Quickfire Questions

17

1. $y = 2x^2 - 3x \rightarrow \frac{dy}{dx}$ ☐
2. $y = 4 - 9x^3 \rightarrow \frac{dy}{dx}$ ☐
3. $y = 5x + 1 \rightarrow$ ☐
4. $y = ax \rightarrow$ ☐
   (where a is a constant)
5. $y = 6x - 3 + px^2 \rightarrow \frac{dy}{dx}$ ☐
   (where p is a constant)

Differentiation

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Harder Example

18

Let $f(x) = 4x^2 - 8x + 3$
a) Find the gradient of $y = f(x)$ at the point $\left(\frac{1}{2}, 0\right)$
b) Find the coordinates of the point on the graph of $y = f(x)$ where the gradient is 8.
c) Find the gradient of $y = f(x)$ at the points where the curve meets the line $y = 4x - 5$.

a

b

c

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Test Your Understanding

19

Let $f(x) = x^2 - 4x + 2$
a) Find the gradient of $y = f(x)$ at the point $(1, -1)$
b) Find the coordinates of the point on the graph of $y = f(x)$ where the gradient is 5.
c) Find the gradient of $y = f(x)$ at the points where the curve meets the line $y = 2 - x$.

a

b

c

Differentiation

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Exercise 12D – Pearson Pure Y1 page 265-266

20

5 Find the gradients of the curve $y = x^2 - 5x + 10$ at the points A and B where the curve meets the line $y = 4$.

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Exercise 12D – Pearson Pure Y1 page 265-266

21

6 Find the gradients of the curve $y = 2x^2$ at the points C and D where the curve meets the line $y = x + 3$.

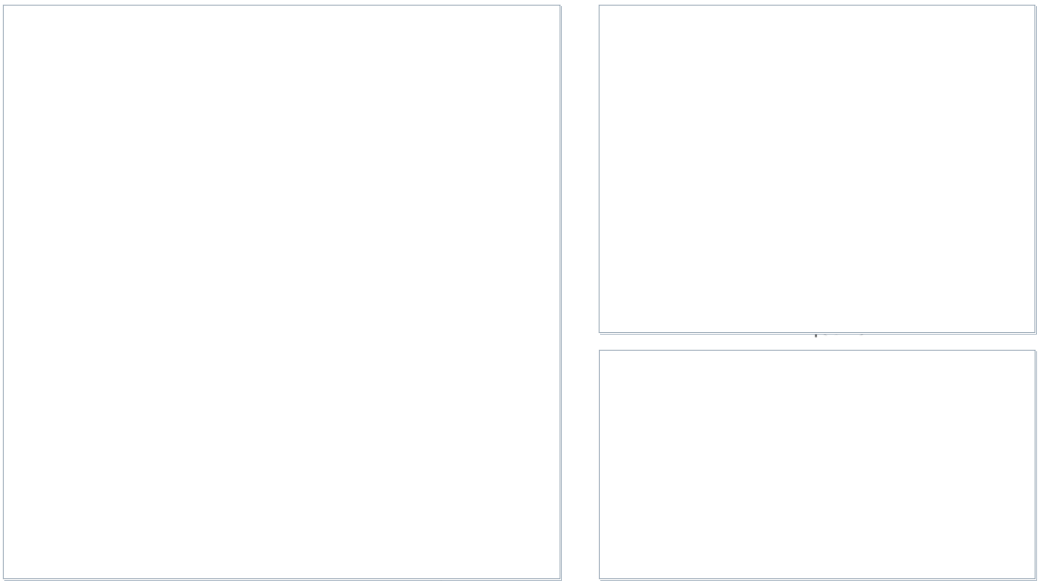
Differentiation

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Exercise 12D – Pearson Pure Y1 page 265-266

22

7\quad $\mathrm{f}(x) = x^2 - 2x - 8$.
a Sketch the graph of $y = \mathrm{f}(x)$.
b On the same set of axes, sketch the graph of $y = \mathrm{f}'(x)$.
c Explain why the x-coordinate of the turning point of $y = \mathrm{f}(x)$ is the same as the x-coordinate of the point where the graph of $y = \mathrm{f}'(x)$ crosses the x-axis.



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Differentiating Harder Expressions

23

If your expression isn't a sum of x^n terms, simply manipulate it until it is!

1. Turn roots into powers:

$y = \sqrt{x} = x$ $\rightarrow$ $\frac{dy}{dx} = \frac{1}{2}$

$y = \frac{1}{\sqrt[3]{x}} = x$ $\rightarrow$ $\frac{dy}{dx} =$

2. Split up fractions.

$y = \frac{x^2+3}{\sqrt{x}} = \frac{x^2+}{x^{\frac{1}{2}}}$

$\rightarrow$ $\frac{dy}{dx} = \frac{3}{2}x^{\frac{1}{2}}$

3. Expand out brackets.

$y = x^2(x-3) =$

$\rightarrow$ $\frac{dy}{dx}$

4. Beware of numbers in denominators!

$y = \frac{1}{3x} =$ $\rightarrow$ $\frac{d}{d}$ x^{-2}

Differentiation

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Test Your Understanding

24

Differentiate the following.

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Exercise 12E – Pearson Pure Y1 page 267-268

25

6\quad $\mathrm{f}(x) = \frac{12}{p\sqrt{x}} + x$, where p is a real constant and $x > 0$. Given that $\mathrm{f}'(2) = 3$, find p, giving your answer in the form $a\sqrt{2}$ where a is a rational number.

Differentiation

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Exercise 12E – Pearson Pure Y1 page 267-268

26

7\quad $\mathrm{f}(x) = (2 - x)^9$.
a Find the first 3 terms, in ascending powers of x, of the binomial expansion of $\mathrm{f}(x)$, giving each term in its simplest form.
b If x is small, so that x^2 and higher powers can be ignored, show that $\mathrm{f}'(x) \approx 9216x - 2304$.

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Extension

27

[MAT 2013 1E]
The expression $\frac{d^2}{dx^2}\left[(2x-1)^4(1-x)^5\right] + \frac{d}{dx}\left[(2x+1)^4(3x^2-2)^2\right]$ is a polynomial of degree:
A) 9
B) 8
C) 7
D) less than 7
E)

This just means “differentiate twice”. The ‘second derivative’ comes later in this chapter.

Full expansion is not needed. The highest power term in the first

Differentiation

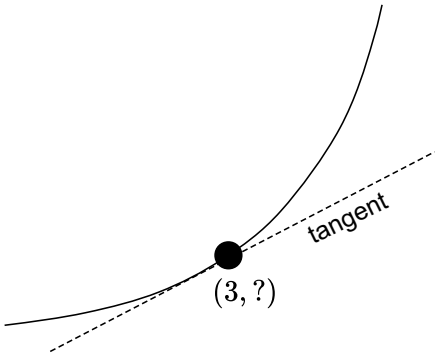
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Finding equations of tangents

28

Find the equation of the **tangent** to the curve $y = x^2$ when $x = 3$.

We want to use $y - y_1 = m\,(x - x_1)$ for the tangent (as it is a straight line!).
Therefore we need:
(a) A point (x_1, y_1)
(b) The gradient m.



Gradient function:

Gradient when $x = 3$:

y-value when $x = 3$:

So equation of tangent:

$$\mathbf{y-9=6\,(x-3)}$$

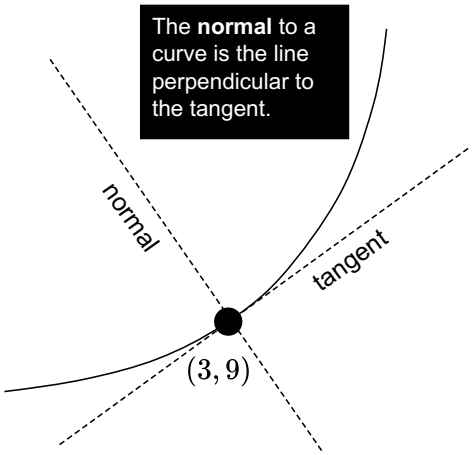
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Finding equations of normals

29

Find the equation of the **normal** to the curve $y = x^2$ when $x = 3$.

The **normal** to a curve is the line perpendicular to the tangent.



Equation of tangent (from earlier):

$$\mathbf{y-9=6(x-3)}$$

Therefore equation of normal:

$$\mathbf{y-9=}\ \tfrac{1}{\mathbf{6}}\mathbf{(x-3)}$$

Differentiation

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Test Your Understanding

30

Find the equation of the **normal** to the curve $y = x + 3\sqrt{x}$ when $x = 9$.

When $x = 9$, $y =$

$x + 3x^{\frac{1}{2}}$

$1 + \frac{3}{2}x^{-\frac{1}{2}}$

Tip: I like to use m_T and m_N to make clear to the examiner (and myself) what gradient I've found.

$= -\frac{2}{3}(x -$

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Exercise 12F – Pearson Pure Y1 page 269-270

31

3 Find the coordinates of the point where the tangent to the curve $y = x^2 + 1$ at the point $(2, 5)$ meets the normal to the same curve at the point $(1, 2)$.

Differentiation

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Exercise 12F – Pearson Pure Y1 page 269-270

32

4 Find the equations of the normals to the curve $y = x + x^3$ at the points $(0, 0)$ and $(1, 2)$, and find the coordinates of the point where these normals meet.

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Exercise 12F – Pearson Pure Y1 page 269-270

33

5 For $\mathrm{f}(x) = 12 - 4x + 2x^2$, find the equations of the tangent and the normal at the point where $x = -1$ on the curve with equation $y = \mathrm{f}(x)$.

Differentiation

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Exercise 12F – Pearson Pure Y1 page 269-270

34

6 The point P with x-coordinate $\frac{1}{2}$ lies on the curve with equation $y = 2x^2$. The normal to the curve at P intersects the curve at points P and Q. Find the coordinates of Q. (6 marks)

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Extension

35

The 'difference of two cubes':
$a^3 - b^3 = (a - b)\left(a^2 + ab + b^2\right)$
will help for both of these.

[STEP I 2005 Q2]
The point P has coordinates $\left(p^2, 2p\right)$ and the point Q has coordinates $\left(q^2, 2q\right)$, where p and q are non-zero and $p \neq q$. The curve C is given by $y^2 = 4x$. The point R is the intersection of the tangent to C at P and the tangent to C at Q. Show that R has coordinates $(pq, p + q)$.

The point S is the intersection of the normal to C at P and the normal to C at Q. If p and q are such that $(1, 0)$ lies on the line PQ, show that S has coordinates $\left(p^2 + q^2 + 1, p + q\right)$, and that the quadrilateral $PSQR$ is a rectangle.

Solutions on next slide.

[STEP I 2012 Q4]
The curve C has equation $xy = \frac{1}{2}$. The tangents to C at the distinct points $P\left(p, \frac{1}{2p}\right)$ and $Q\left(q, \frac{1}{2q}\right)$, where p and q are positive, intersect at T and the normal to C at these points intersect at N. Show that T is the point $\left(\frac{2pq}{p+q}, \frac{1}{p+q}\right)$
In the case $pq = \frac{1}{2}$, find the coordinates of N. Show (in this case) that T and N lie on the line $y = x$ and are such that the product of their distances from the origin is constant.

Differentiation

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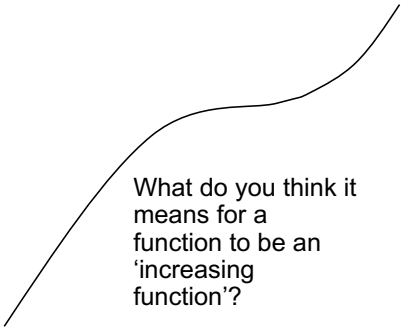
Solutions to Extension Question 1

36

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Increasing and Decreasing Functions

37

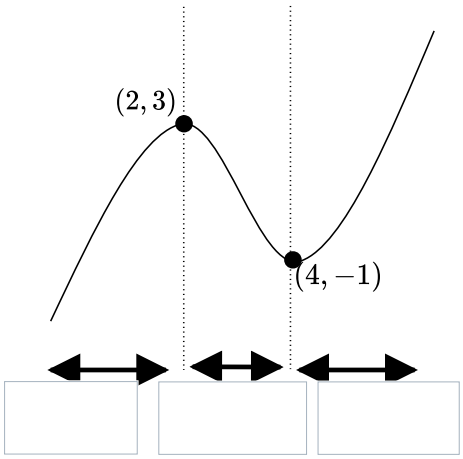


What do you think it means for a function to be an 'increasing function'?

✎ An increasing function is one whose gradient is always at least 0.
$f'(x) \geq 0$ for all x.

It would be '**strictly increasing**' if $f(x) > 0$ for all x, i.e. is not allowed to go horizontal.

A function can also be increasing and decreasing in certain intervals.



We could also write "$f(x)$ is decreasing in the interval $[2, 4]$"
$[a,b]$ represents all the real numbers between a and b inclusive, i.e:

$$[a, b] = \{x : a \leq x \leq b\}$$

Differentiation

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Examples

38

Show that the function $f(x) = x^3 + 6x^2 + 21x + 2$ is increasing for all real values of x.

Find the interval on which the function $f(x) = x^3 + 3x^2 - 9x$ is decreasing.

Tip: To show a quadratic is always positive, complete the square, then indicate the squared term is always at least 0.

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Test Your Understanding

39

Show that the function $f(x) = x^3 + 16x - 2$ is increasing for all real values of x.

Find the interval on which the function $f(x) = x^3 + 6x^2 - 135x$ is decreasing.

Differentiation

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Exercise 12G – Pearson Pure Y1 page 271

40

3 Show that the function $\mathrm{f}(x) = 4 - x(2x^2 + 3)$ is decreasing for all $x \in \mathbb{R}$.

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Exercise 12G – Pearson Pure Y1 page 271

41

4 a Given that the function $\mathrm{f}(x) = x^2 + px$ is increasing on the interval $[-1, 1]$, find one possible value for p.
b State with justification whether this is the only possible value for p.

Differentiation

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Second Order Derivatives

42

When you differentiate once, the expression you get is known as the **first derivative**. Unsurprisingly, when we differentiate a second time, the resulting expression is known as the **second derivative**. And so on…

Original Function	First Derivative	Second Derivative
$y = x^4$ (Leibniz's)	$\frac{dy}{dx} = 4x^3$	$\frac{d^2y}{dx^2} = 12x^2$
$y = x^4$	$y\prime = 4x^3$	$y\prime\prime = 12x^2$
$y = x^4$ (Newton's)	$\dot{y} = 4x^3$	$\ddot{y} = 12x^2$
$f(x) = x^4$ (Lagrange's)	$f\prime(x) = 4x^3$	$f\prime\prime(x) = 12x^2$

You can similarly have the third derivative ($\frac{d^3y}{dx^3}$), although this is no longer in the A Level syllabus. We'll see why might use the second derivative soon…

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Just for your interest…

43

How does the notation $\frac{d^2y}{dx^2}$ work?
Why are the squareds where they are?

Suppose that $y = x^3 + 1$.
Then when we write $\frac{dy}{dx}$, we're effectively doing $\frac{d(x^3+1)}{dx}$ (by substitution), although this would typically be written:

$$\frac{d}{dx}\left(x^3 + 1\right)$$

The $\frac{d}{dx}(\dots)$ notation is quite handy, because it behaves as a function and allows us to write the original expression and the derivative within a single equation:

$$\frac{d}{dx}\left(x^3 + 1\right) = 3x^2$$

Therefore, if we wanted to differentiate y twice, we'd do:

$$\frac{d}{dx}\left(\frac{d}{dx}(y)\right)$$
$$= \frac{d^2}{dx^2}(y)$$
$$= \frac{d^2y}{dx^2}$$

Differentiation

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Examples

44

If $y = 3x^5 + \frac{4}{x^2}$, find $\frac{d^2y}{dx^2}$.

If $f(x) = 3\sqrt{x} + \frac{1}{2\sqrt{x}}$, find $f''(x)$.

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Test Your Understanding

45

If $y = 5x^3 - \frac{x}{3\sqrt{x}}$, find $\frac{d^2y}{dx^2}$.

Differentiation

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Exercise 12H – Pearson Pure Y1 page 272

46

3 Given that $y = (2x - 3)^3$, find the value of x when $\dfrac{\mathrm{d}^2y}{\mathrm{d}x^2} = 0$.

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Exercise 12H – Pearson Pure Y1 page 272

47

4\quad $\mathrm{f}(x) = px^3 - 3px^2 + x^2 - 4$. When $x = 2$, $\mathrm{f}''(x) = -1$. Find the value of p.

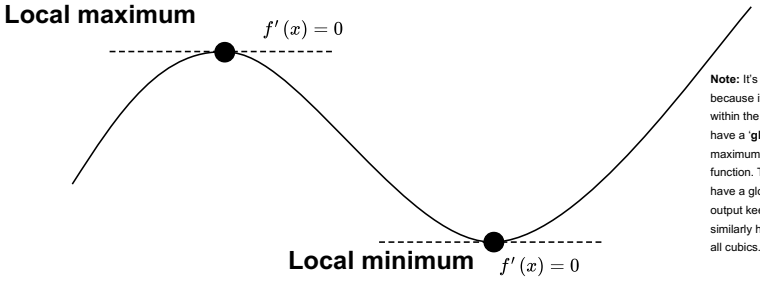
Differentiation

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Stationary/Turning Points

48

A stationary point is where the gradient is 0, i.e. $f'(x) = 0$.



Note: It's called a '**local**' maximum because it's the function's largest output within the vicinity. Functions may also have a '**global**' maximum, i.e. the maximum output across the entire function. This particular function doesn't have a global maximum because the output keeps increasing up to infinity. It similarly has no global minimum, as with all cubics.

Find the coordinates of the turning points of $y = x^3 + 6x^2 - 135x$

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More Examples

49

Find the least value of
$f(x) = x^2 - 4x + 9$

Note: Method 2 is only applicable for quadratic functions. For others, differentiation must be used.

Find the turning point of
$y = \sqrt{x} - x$

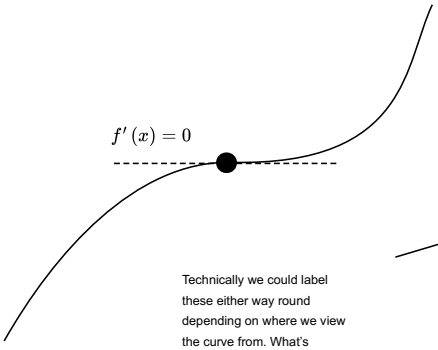
Differentiation

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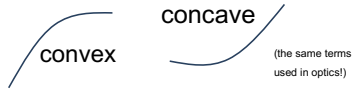
Points of Inflection

50

There's a third type of stationary point (that we've encountered previously):



A point of inflection is where the curve changes from convex concave (or vice versa).

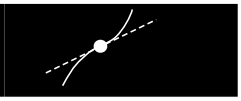


(the same terms used in optics!)

Technically we could label these either way round depending on where we view the curve from. What's important is that the **concavity changes**.

i.e. the line curves in one direction before the point of inflection, then curves in the other direction after.

Side Note: Not all points of inflection are stationary points, as can be seen in the example on the right.
A point of inflection which **is** a stationary point is known as a *saddle point*.

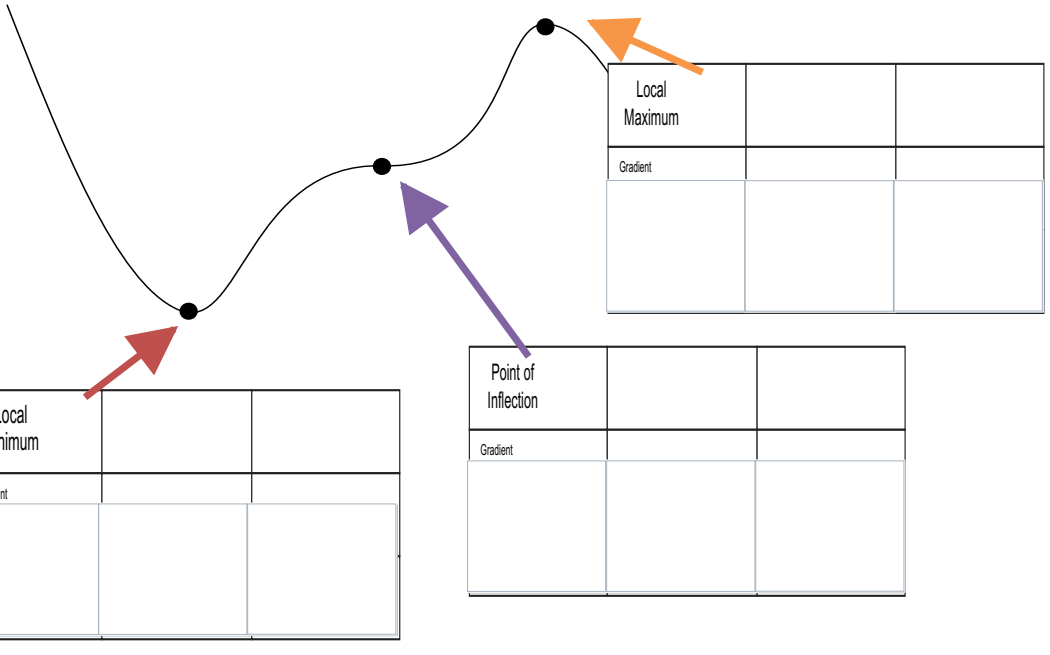


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How do we tell what type of stationary point?

51

Method 1: Look at gradient just before and just after point.



Differentiation

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How do we tell what type of stationary point?

52

Method 1: Look at gradient just before and just after point.

Find the stationary point on the curve with equation $y = x^4 - 32x$, and determine whether it is a local maximum, a local minimum or a point of inflection.

Strategy: Find the gr

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Method 2: Using the second derivative

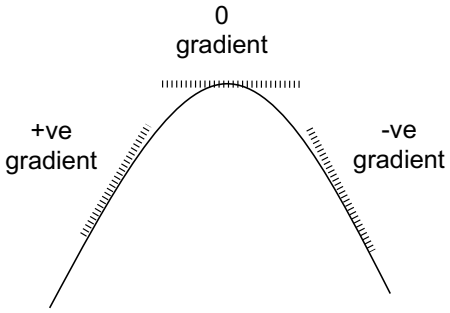
53

The method of substituting values of x just before and after is a bit cumbersome.
It also has the potential for problems: what if two different types of stationary points are really close together?

Recall the gradient gives a measure of the **rate of change** of y, i.e. how much the y value changes as x changes.

Thus by differentiating the gradient function, the **second derivative tells us the rate at what the gradient is changing**.

Thus if the second derivative is positive, the gradient is increasing.
If the second derivative is negative, the gradient is decreasing.



At a maximum point, we can see that as x increases, the gradient is decreasing from a positive value to a negative value.

$$\therefore \frac{d^2y}{dx^2} < 0$$

Differentiation

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Method 2: Using the second derivative

54

✎ At a stationary point $x = a$:
• If $f''(a) > 0$ the point is a local minimum.
• If $f''(a) < 0$ the point is a local maximum.
• If $f''(a) = 0$ it could be any type of point, so resort to Method 1.

The stationary point of $y = x^4 - 32x$ is $(2, -48)$. Use the second derivative to classify this stationary point.

I will eventually do a 'Just for your Interest…' thingy on why we can't classify the point when $f''(x) = 0$, and how we could use the **third derivative**!

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Test Your Understanding

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Edexcel C2 May 2013 Q9

The curve with equation $y = x^2 - 32\sqrt{x} + 20$, $x > 0$, has a stationary point P. Use calculus
(a) to find the coordinates of P, (6)
(b) to determine the nature of the stationary point P. (3)

(a)
$2x$
obt
[or
$(x^3$
$x =$
A1,

(b)
$\left(\frac{d^2}{da}\right.$

Differentiation

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Sketching Graphs

56

All the way back in Chapter 4, we used features such as intercepts with the axes, and behaviour when $x \to \infty$ and $x \to -\infty$ in order to sketch graphs. Now we can also find stationary/turning points!

[Textbook] By first finding the stationary points, sketch the graph of $y = \frac{1}{x} + 27x^3$

? As $x \to \infty$, $x \to -\infty$

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Exercise 12I – Pearson Pure Y1 page 276

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5 By considering the gradient on either side of the stationary point on the curve $y = x^3 - 3x^2 + 3x$, show that this point is a point of inflection. Sketch the curve $y = x^3 - 3x^2 + 3x$.

Differentiation

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Exercise 12I – Pearson Pure Y1 page 276

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6 Find the maximum value and hence the range of values for the function $\mathrm{f}(x) = 27 - 2x^4$.

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7\quad $\mathrm{f}(x) = x^4 + 3x^3 - 5x^2 - 3x + 1$.
a Find the coordinates of the stationary points of $\mathrm{f}(x)$, and determine the nature of each.
b Sketch the graph of $y = \mathrm{f}(x)$.

Differentiation

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Extension

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1 [MAT 2014 1C] The cubic $y = kx^3 - (k+1)x^2 + (2-k)x - k$ has a turning point, that is a minimum, when $x = 1$ precisely for
A) $k > 0$
B) $0 < k < 1$
C) $k > \frac{1}{2}$
D) $k < 3$
E) all values of k
F)

2 *[MAT 2004 1B]* The smallest value of the function:
$f(x) = 2x^3 - 9x^2 + 12x + 3$
In the range $0 \leq x \leq 2$ is what?

3 *[MAT 2001 1E]* The maximum gradient of the curve $y = x^4 - 4x^3 + 4x^2 + 2$ in the range $0 \leq x \leq 2\frac{1}{5}$ occurs when:
A) $x = 0$
B) $x = 1 - \frac{1}{\sqrt{3}}$
C) $x = 1 + \frac{1}{\sqrt{3}}$
D) $x = 2\frac{1}{5}$
E)

Hint: When two curves touch, their y values must match, but what else must also match?

[STEP I 2007 Q8] A curve is
4 $y = ax^3 - 6ax^2 + (12a+12)x - (8a+16)$
where a is a real number. Show that this curve touches the curve with equation $y = x^3$ at $(2, 8)$. Determine the coordinates of any other point of intersection of the two curves.
(i) Sketch on the same axes these two curves when $a = 2$.
(ii) … when $a = 1$ (iii) when $a = -2$

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Solutions to Extension Questions

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Differentiation

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Solutions to Extension Questions

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Solutions to Extension Questions

63

Differentiation

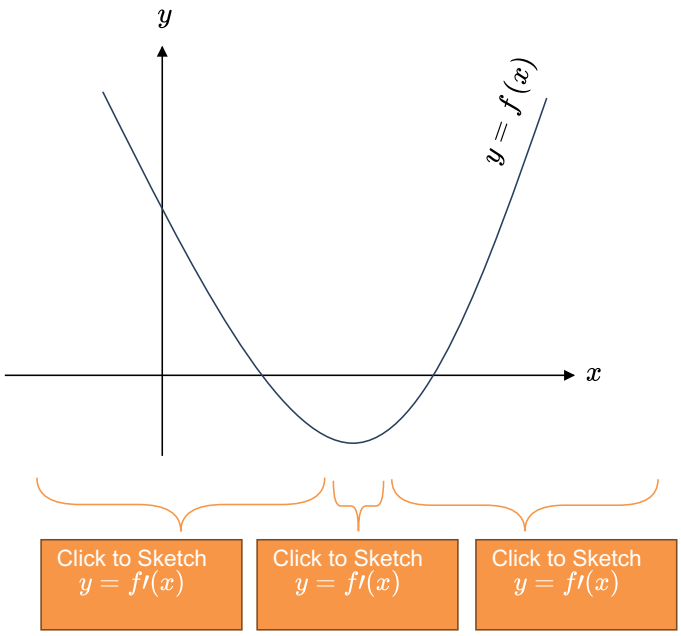
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Sketching Gradient Functions

64

The new A Level specification specifically mentions being able to sketch $y = f\prime(x)$.

If you know the function $f\prime(x)$ explicitly (e.g. because you differentiated $y = f(x)$), you can use your knowledge of sketching straight line/ quadratic/cubic graphs. But in other cases **you won't be given the function explicitly**, but **just the sketch**.

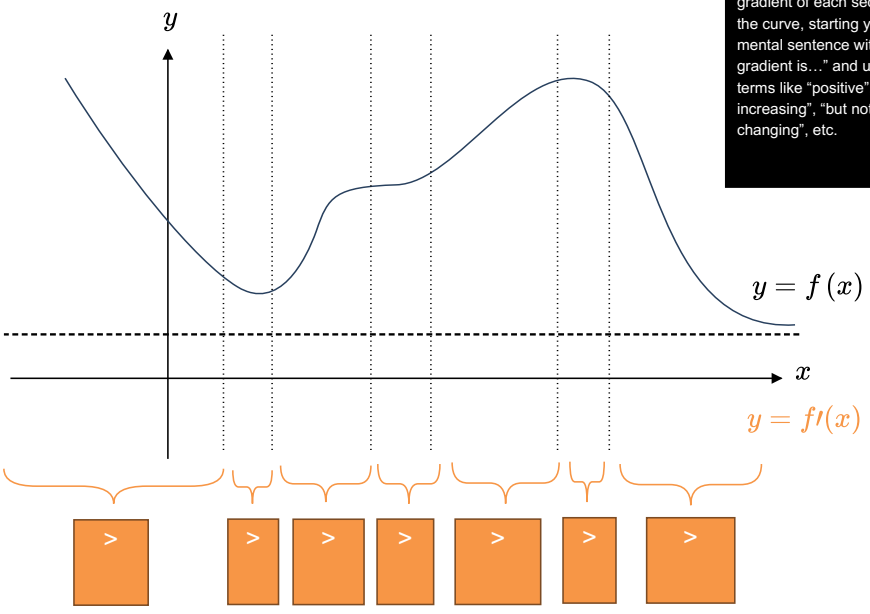


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A Harder One

65

Tip: Mentally describe the gradient of each section of the curve, starting your mental sentence with "The gradient is…" and using terms like "positive", "but increasing", "but not changing", etc.

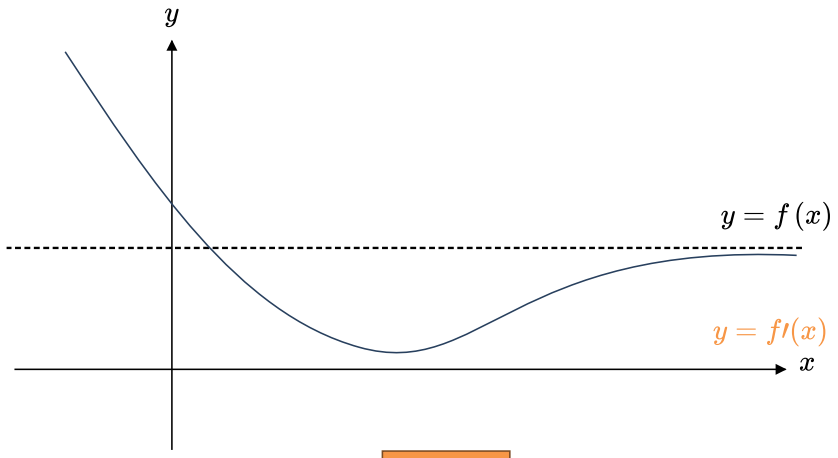


Differentiation

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Test Your Understanding

66



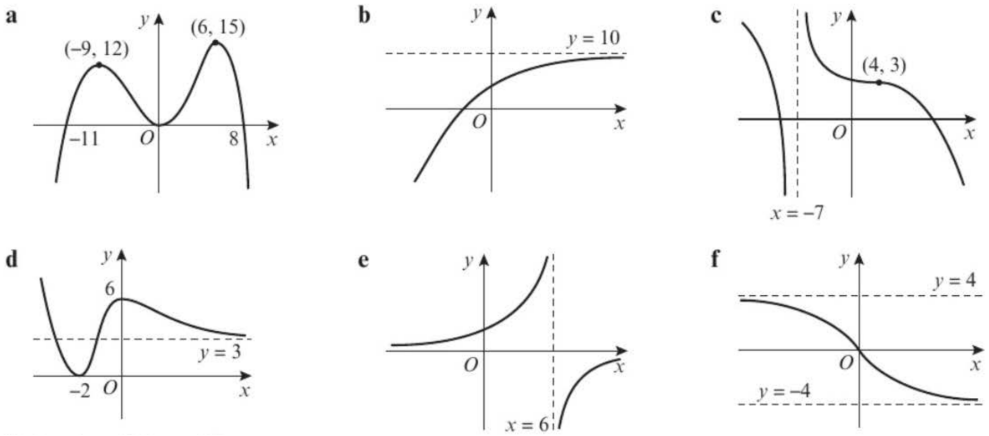
Solution >

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67

1 For each graph given, sketch the graph of the corresponding gradient function on a separate set of axes. Show the coordinates of any points where the curve cuts or meets the x-axis, and give the equations of any asymptotes.



Solutions here

Differentiation

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2\quad $\mathrm{f}(x) = (x+1)(x-4)^2$.
a Sketch the graph of $y = \mathrm{f}(x)$.
b On a separate set of axes, sketch the graph of $y = \mathrm{f}'(x)$.
c Show that $\mathrm{f}'(x) = (x-4)(3x-2)$.
d Use the derivative to determine the exact coordinates of the points where the gradient function cuts the coordinate axes.

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Extension

69

1 [MAT 2015 1B]

$$f(x) = (x+a)^n$$

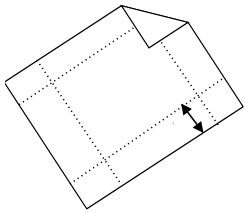
where a is a real number and n is a positive whole number, and $n \geq 2$. If $y = f(x)$ and $y = f\prime(x)$ are plotted on the same axes, the number of intersections between $f(x)$ and $f'(x)$ will:
A) always be odd
B) always be even
C) depend on a but not n
D) depend on n but not a
E) depend on both a and n.
F)

Differentiation

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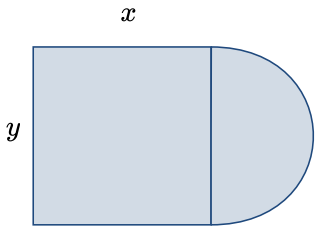
Optimisation Problems/Modelling

70



We have a sheet of A4 paper, which we want to fold into a cuboid. What height should we choose for the cuboid in order to maximise the volume?

These are examples of **optimisation problems**: we're trying to maximise/minimise some quantity by choosing an appropriate value of a variable that we can control.



We have 50m of fencing, and want to make a bear pen of the following shape, such that the area is maximised. What should we choose x and y to be?

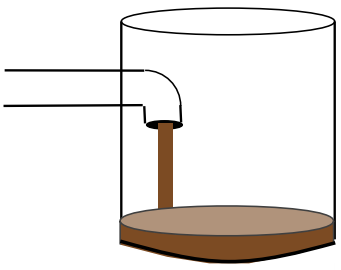
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'Rate of change'

71

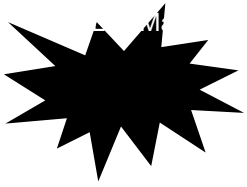
Up to now we've had y in terms of x, where $\frac{dy}{dx}$ means "the **rate** at which y changes with respect to x".

But we can use similar gradient function notation for other **physical quantities**.



A sewage container fills at a rate of 20 cm^3 per second.

How could we use appropriate notation to represent this?



Differentiation

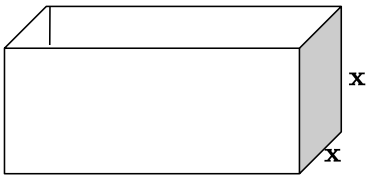
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Example Optimisation Problem

72

Optimisation problems in an exam usually follow the following pattern:

- There are 2 variables involved (you may have to introduce one yourself), typically lengths.
- There are expressions for **two different physical quantities**:
  - One is a **constraint**, e.g. "the surface area is 20cm^2".
  - The **other we wish to maximise/minimise**, e.g. "we wish to maximise the volume".
- We use the constraint to **eliminate one of the variables** in the latter equation, so that it is then **just in terms of one variable**, and we can then use differentiation to find the turning point.



[Textbook] A large tank in the shape of a cuboid is to be made from 54m^2 of sheet metal. The tank has a horizontal base and no top. The height of the tank is x metres. Two of the opposite vertical faces are squares.

a) Show that the volume, V m^3, of the tank is given by $V = 18x - \frac{2}{3}x^3$.

b) Given that x can vary, use differentiation to find the maximum or minimum value of V.

$\frac{dV}{dt}$

$V =$

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Test Your Understanding

73

Edexcel C2 May 2011 Q8

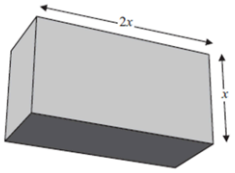


Figure 2

A cuboid has a rectangular cross-section where the length of the rectangle is equal to twice its width, x cm, as shown in Figure 2.

The volume of the cuboid is 81 cubic centimetres.

(a) Show that the total length, L cm, of the twelve edges of the cuboid is given by $L = 12x + \frac{162}{x^2}$.

\hfill (3)

(b) Use calculus to find the minimum value of L. \hfill (6)

(c) Justify, by further differentiation, that the value of L that you have found is a minimum. \hfill (2)

(b) $\frac{dL}{dx}$

Either 1

Correct $\{\frac{dL}{dx} =$

$L' = 0$

$x = \sqrt[3]{2}$

$\{x = 3$

Substitu

$54 \qua$

(c) {For

Correct f

$\frac{972}{x^3}$ an

(a)

$\{V =\}$

$\{L = 2($

$y = \frac{81}{2x^2}$

Making y

So, $L =$

Correct s

Differentiation

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Exercise 12K – Pearson Pure Y1 page 81

74

6 A rectangular garden is fenced on three sides, and the house forms the fourth side of the rectangle.

a Given that the total length of the fence is $80\,\text{m}$, show that the area, A, of the garden is given by the formula $A = y(80 - 2y)$, where y is the distance from the house to the end of the garden.

b Given that the area is a maximum for this length of fence, find the dimensions of the enclosed garden, and the area which is enclosed.

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7 A closed cylinder has total surface area equal to 600π.
a Show that the volume, $V\,\text{cm}^3$, of this cylinder is given by the formula $V = 300\pi r - \pi r^3$, where $r\,\text{cm}$ is the radius of the cylinder.
b Find the maximum volume of such a cylinder.

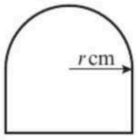
Differentiation

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Exercise 12K – Pearson Pure Y1 page 81

76

9 A shape consists of a rectangular base with a semicircular top, as shown.



a Given that the perimeter of the shape is $40\,\text{cm}$, show that its area, $A\,\text{cm}^2$, is given by the formula

$A = 40r - 2r^2 - \frac{\pi r^2}{2}$

where r cm is the radius of the semicircle. \hfill (2 marks)

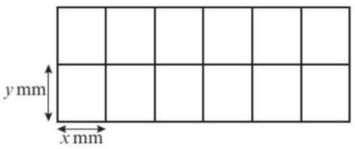
b Hence find the maximum value for the area of the shape. \hfill (4 marks)

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77

10 The shape shown is a wire frame in the form of a large rectangle split by parallel lengths of wire into 12 smaller equal-sized rectangles.



a Given that the total length of wire used to complete the whole frame is $1512\,\text{mm}$, show that the area of the whole shape, $A\,\text{mm}^2$, is given by the formula

$A = 1296x - \frac{108x^2}{7}$

where x mm is the width of one of the smaller rectangles. \hfill (4 marks)

b Hence find the maximum area which can be enclosed in this way. \hfill (4 marks)